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Studierichting: Informatica
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$$\begin{aligned} & \int \frac{\sqrt{x^2 - 1}}{x} dx \\ &= \int \frac{\sqrt{x^2 - 1}}{x^2} x dx \\ & t = \sqrt{x^2 - 1} \Rightarrow t^2 = x^2 - 1 \Rightarrow 2t dt = 2x dx \Rightarrow t dt = x dx \\ &= \int \frac{t}{t^2 + 1} t dt \\ &= \int \frac{t^2}{t^2 + 1} dt \\ &= \int \frac{t^2 + 1 - 1}{t^2 + 1} dt \\ &= \int \frac{t^2 + 1}{t^2 + 1} dt - \int \frac{1}{t^2 + 1} dt \\ &= \int 1 dt - \int \frac{1}{t^2 + 1} dt \\ &= t - \text{Bgtan } t + C \\ &= \sqrt{x^2 - 1} - \text{Bgtan } \sqrt{x^2 - 1} + C \end{aligned}$$

References